

Inverse Functions and Writing Equations Given Their Graphs

General Forms of Trigonometric Functions

Sinusoidal
Functions.

$$y = A \sin(\omega x - \phi) + B$$

$$y = A \cos(\omega x - \phi) + B$$

$$y = A \csc(\omega x - \phi) + B$$

$$y = A \tan(\omega x - \phi) + B$$

$$y = A \sec(\omega x - \phi) + B$$

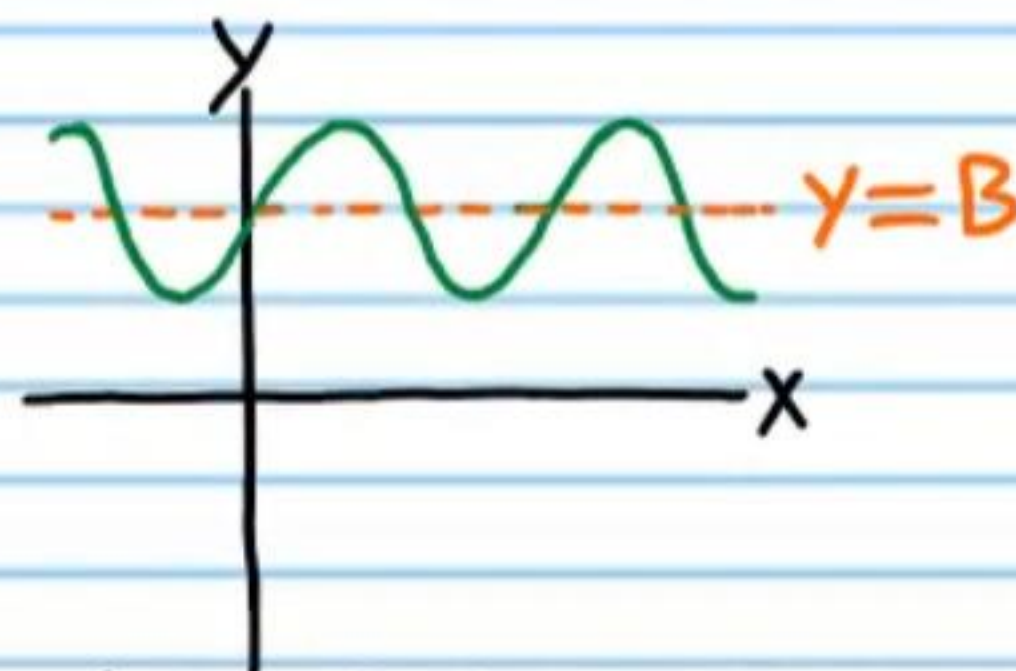
$$y = A \cot(\omega x - \phi) + B$$

Sinusoidal Function: A function whose graph has the same general form as the sine or cosine functions.

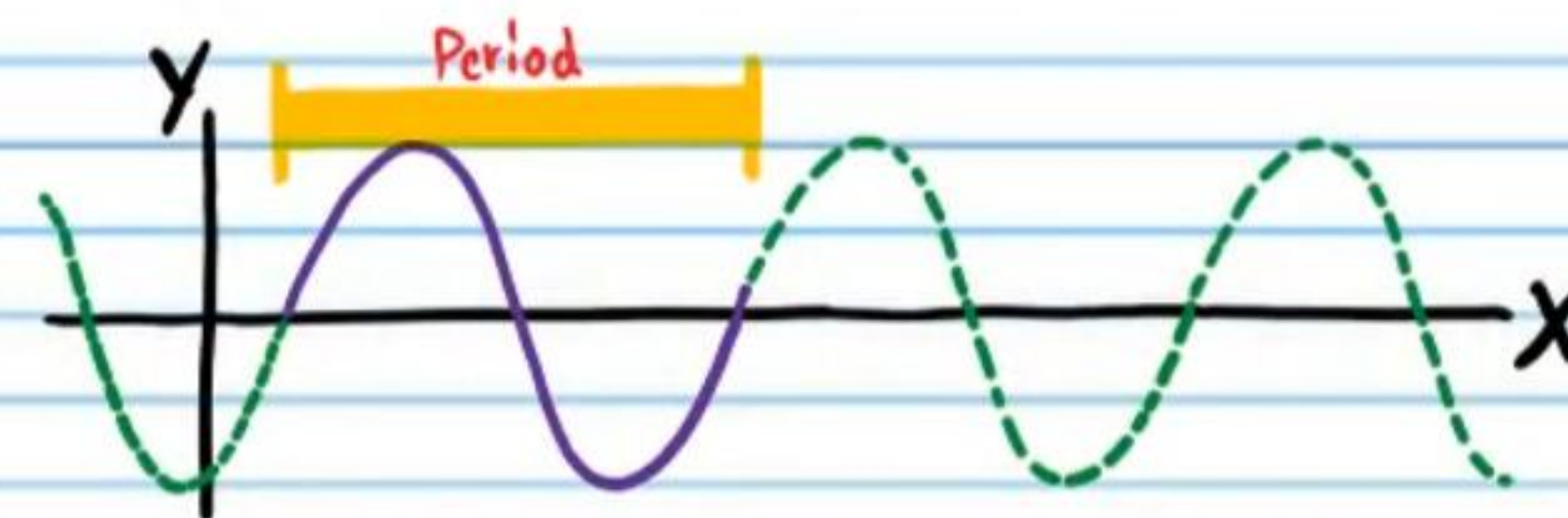
Midline: A horizontal line drawn through the middle of the graph of a trigonometric function to assist when graphing.

For the functions above, the equation of the midline is $y = B$.

Example:



Period: The number of degrees required for a trigonometric function to complete one whole cycle.



The period of the sine, cosine, cosecant, and secant functions is given by the following equation.

$$P = \frac{360^\circ}{\omega}$$

The period of the tangent and cotangent functions is given by the following equation.

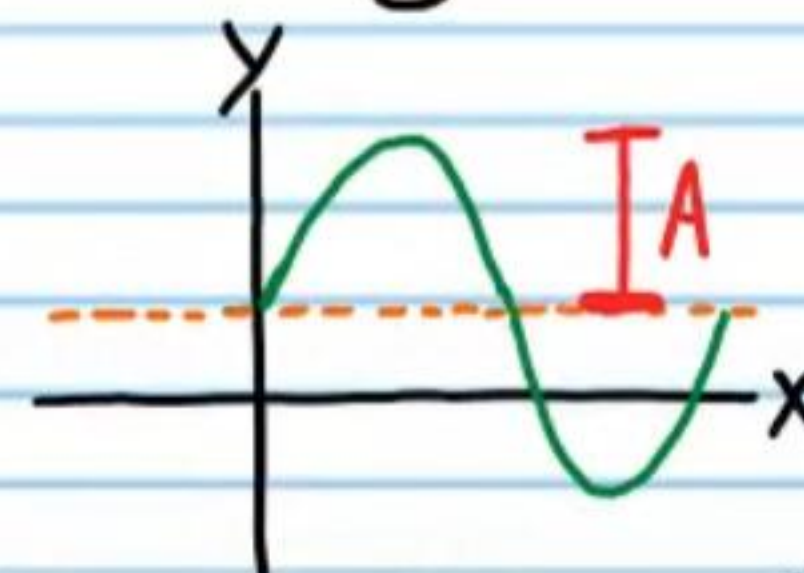
$$P = \frac{180^\circ}{\omega}$$

Phase Shift: The horizontal distance that a parent graph is translated (to the right or left).

The phase shift is given by the equation below

$$PS = \frac{\phi}{\omega}$$

Amplitude: The vertical distance from the midline to the highest or lowest point on a sinusoidal graph.



The amplitude is the magnitude (i.e. the absolute value) of the coefficient of the trigonometric function.

Example: If $y = A \sin(\omega x - \phi) + B$, then Amplitude = $|A|$.

Writing Equations Given Their Graphs

Write an equation corresponding to each graph.

★ There is more than one equation that can be written for each graph (e.g. cosine functions or sine functions can represent the same graph). To avoid confusion, use the sine or cosecant functions rather than the cosine or secant functions. ★ Each horizontal unit on the graphs is 90° . Each vertical unit is one.

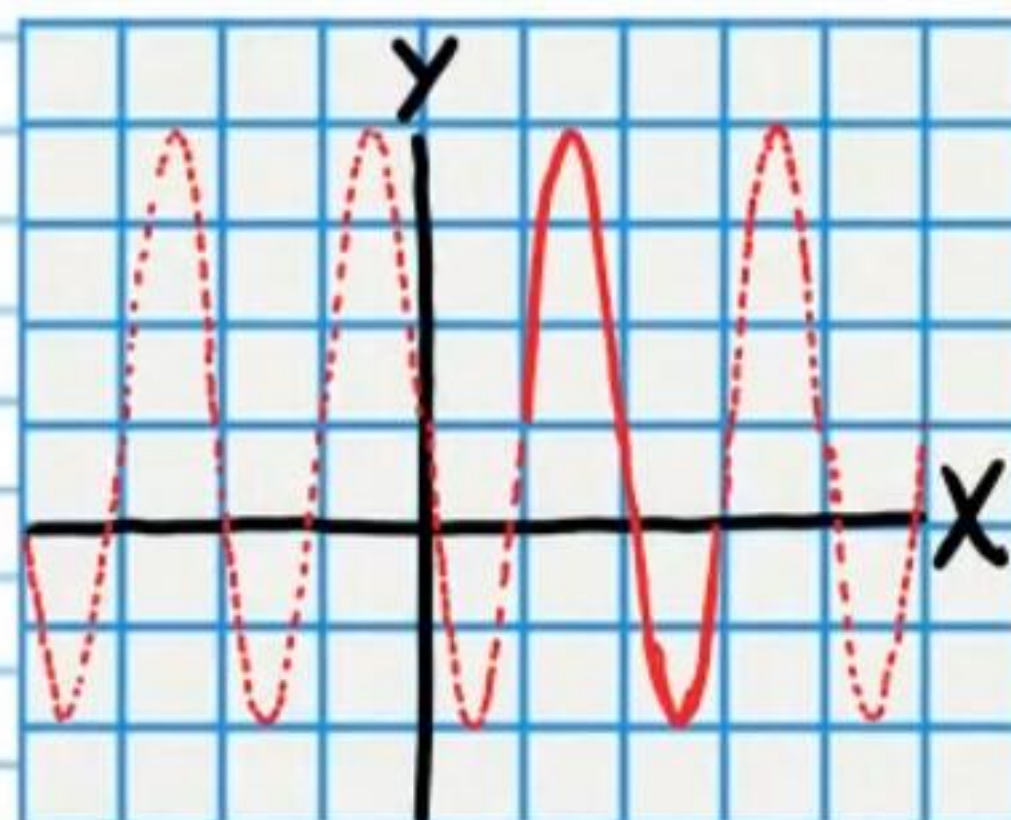
General Form of the Sine Function: $y = A \sin(\omega x - \phi) + B$

- ① Midline Equation: $y = 1$

$$B = 1$$

$$\text{Amplitude: } A = 3$$

$$\text{Omega: } P = \frac{360^\circ}{\omega}$$



Phase Shift:

$$PS = \frac{\phi}{\omega}$$

$$90^\circ = \frac{\phi}{2}$$

$$180^\circ = \phi$$

$$180^\circ = \frac{360^\circ}{\omega}$$

$$\omega \cdot 180^\circ = \frac{360^\circ}{\omega} \cdot \omega$$

$$180^\circ \omega = 360^\circ$$

$$\omega = 2$$

$$\boxed{y = 3 \sin(2x - 180^\circ) + 1}$$

General Form of the Cosecant Function: $y = A \csc(\omega x - \phi) + B$

- ② Midline Equation: $y = 0$

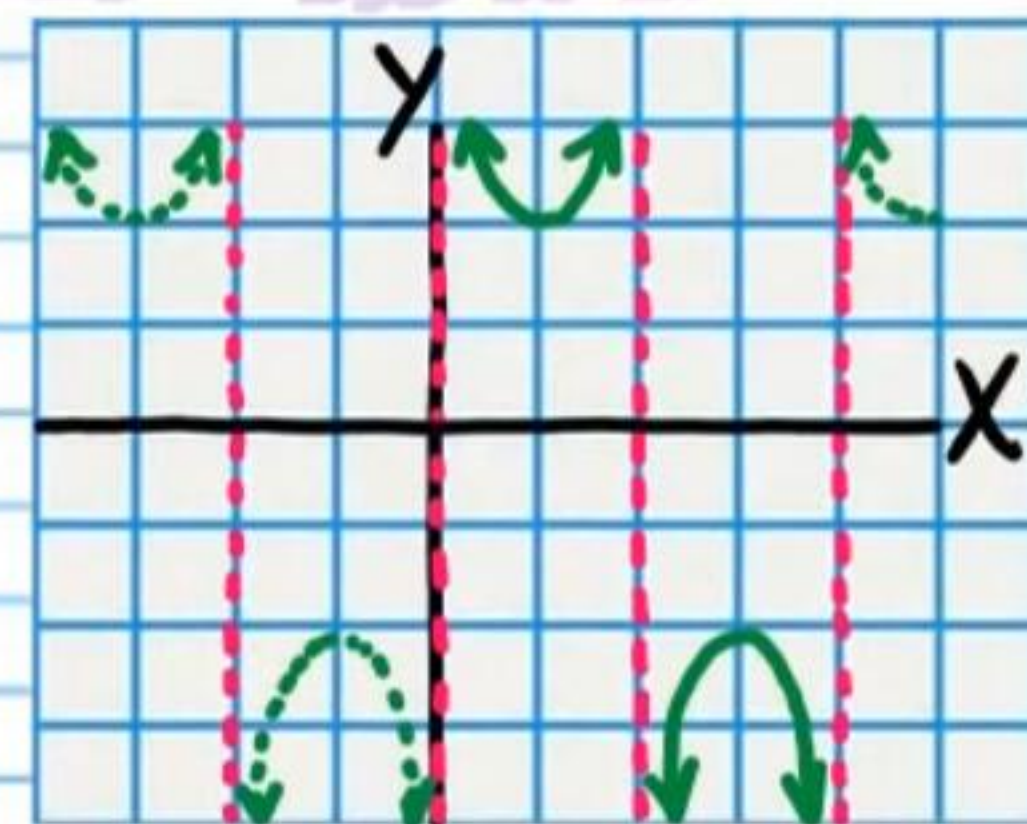
$$B = 0$$

$$A: A = 2$$

$$\text{Omega: } P = \frac{360^\circ}{\omega}$$

$$360^\circ = \frac{360^\circ}{\omega}$$

$$\omega = 1$$



$$y = 2 \csc(1 \cdot x - 0) + 0$$

$$\boxed{y = 2 \csc x}$$

$$\text{Phase Shift: } PS = \frac{\phi}{\omega}$$

$$0 = \frac{\phi}{1}$$

$$0 = \phi$$

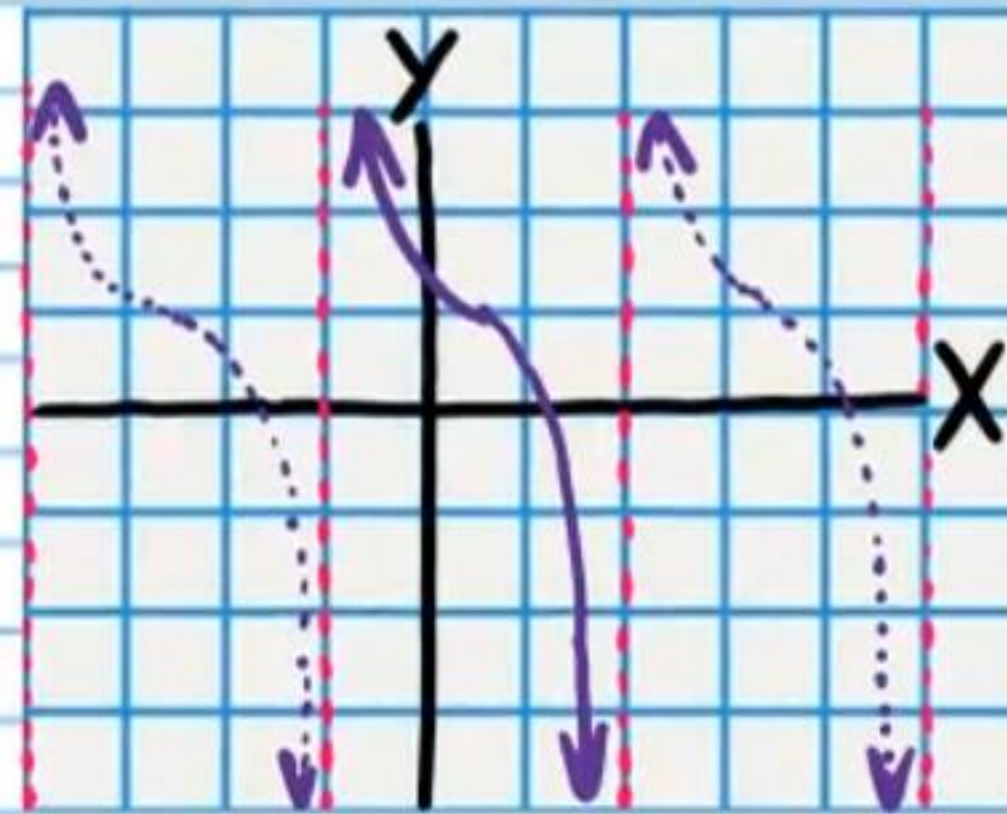
General Form of the Cotangent Function: $y = A \cot(Wx - \phi) + B$

③ Midline Equation: $y = 1$

$$B = 1$$

A: $\star$ Assume $A = 1$

$$\text{Omega: } P = \frac{180^\circ}{W}$$



$$270^\circ = \frac{180^\circ}{W}$$

$$W \cdot 270^\circ = \frac{180^\circ}{W} \cdot W$$

$$270^\circ W = 180^\circ$$

$$W = \frac{180^\circ}{270^\circ}$$

$$W = \frac{2}{3}$$

$$\text{Phase Shift: } PS = \frac{\phi}{W}$$

$$-90^\circ = \frac{\phi}{\frac{2}{3}}$$

$$\frac{2}{3}(-90^\circ) = \frac{\phi}{\frac{2}{3}} \cdot \frac{2}{3}$$

$$-60^\circ = \phi$$

$$y = 1 \cdot \cot\left(\frac{2}{3}x - (-60^\circ)\right) + 1$$

$$y = \cot\left(\frac{2}{3}x + 60^\circ\right) + 1$$

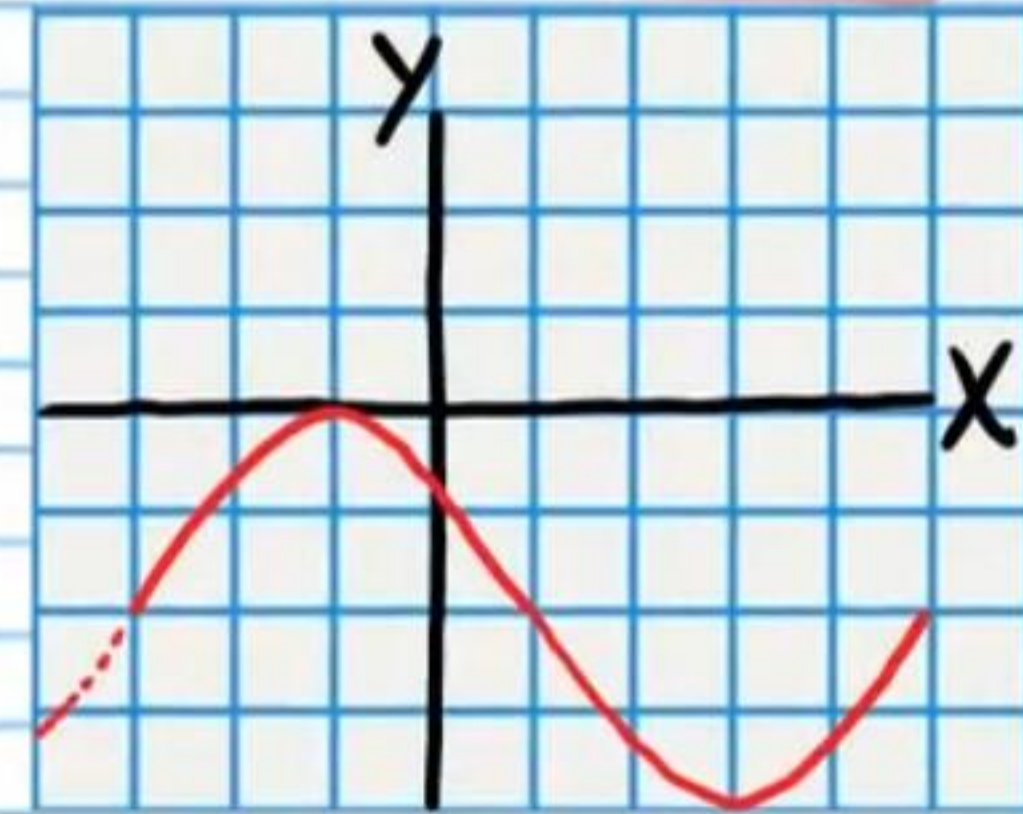
General Form of the Sine Function: $y = A \sin(Wx - \phi) + B$

④ Midline Equation: $y = -2$

$$B = -2$$

Amplitude: $A = 2$

$$\text{Omega: } P = \frac{360^\circ}{W}$$



$$720^\circ = \frac{360^\circ}{W}$$

$$W \cdot 720^\circ = \frac{360^\circ}{W} \cdot W$$

$$720^\circ W = 360^\circ$$

$$W = \frac{360^\circ}{720^\circ}$$

$$W = \frac{1}{2}$$

$$\text{Phase Shift: } PS = \frac{\phi}{W}$$

$$PS = \frac{\phi}{\frac{1}{2}}$$

$$-270^\circ = \frac{\phi}{\frac{1}{2}}$$

$$\frac{1}{2}(-270^\circ) = \frac{\phi}{\frac{1}{2}} \cdot \frac{1}{2}$$

$$-135^\circ = \phi$$

$$y = 2 \sin\left(\frac{1}{2}x - (-135^\circ)\right) + (-2)$$

$$y = 2 \sin\left(\frac{1}{2}x + 135^\circ\right) - 2$$

General Form of the Cosecant Function: $y = A \csc(Wx - \phi) + B$

⑤ Midline Equation: $y = 3$

$$B = 3$$

$$A: A = 1$$

$$\text{Omega: } P = \frac{360^\circ}{W}$$

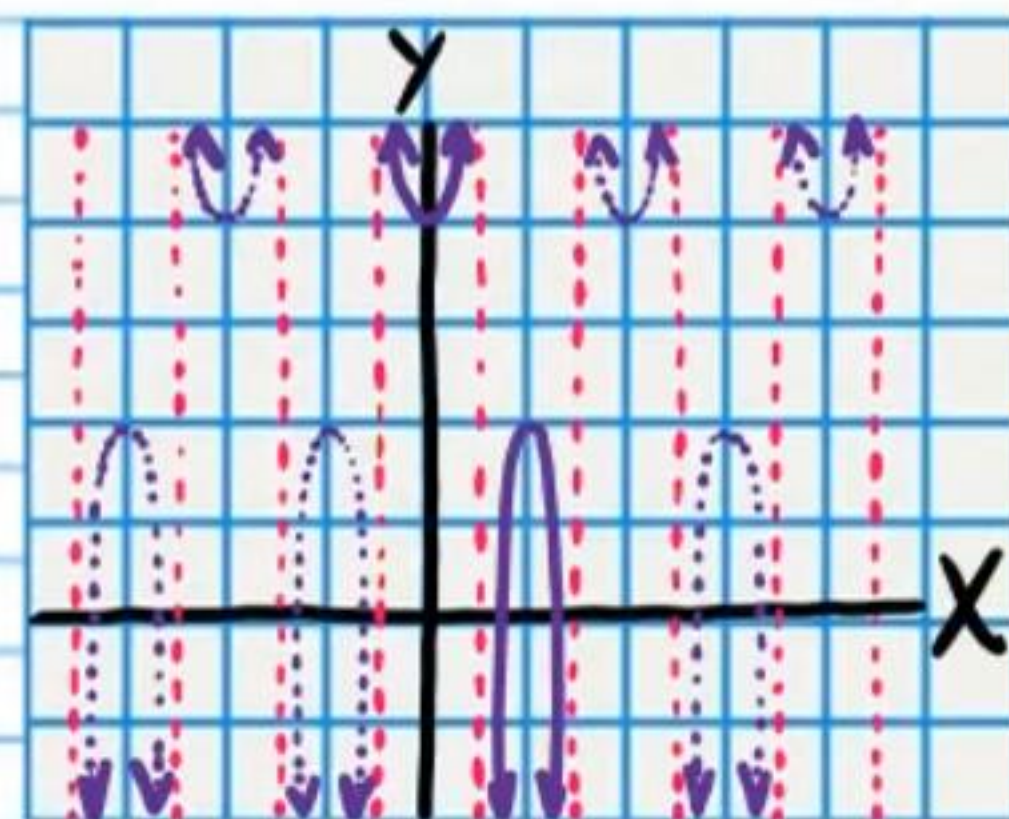
$$180^\circ = \frac{360^\circ}{W}$$

$$W = 2$$

$$\text{Phase Shift: } PS = \frac{\phi}{W}$$

$$-45^\circ = \frac{\phi}{2}$$

$$-90^\circ = \phi$$



$$y = (1) \csc(2x - (-90^\circ)) + 3$$

$$\boxed{y = \csc(2x + 90^\circ) + 3}$$

General Form of the tangent Function: $y = A \tan(Wx - \phi) + B$

⑥ Midline Equation: $y = -2$

$$B = -2$$

$$A: \star \text{ Assume } A = 1.$$

$$\text{Omega: } P = \frac{180^\circ}{W}$$

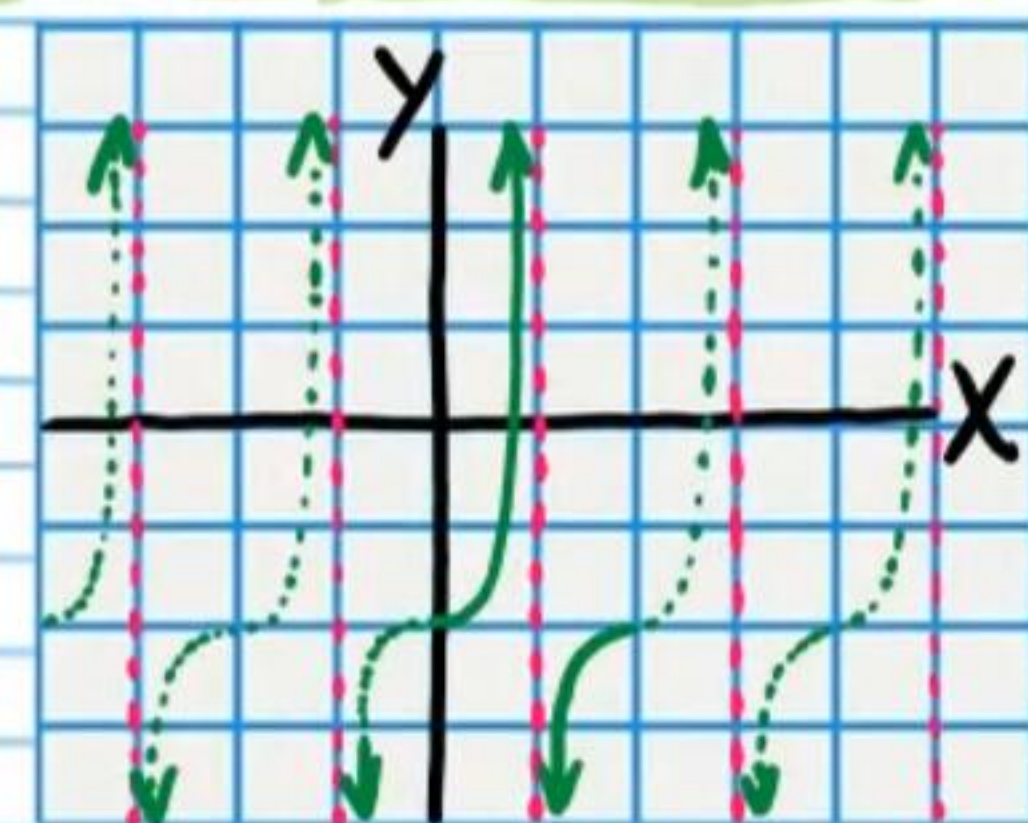
$$180^\circ = \frac{180^\circ}{W}$$

$$W = 1$$

$$\text{Phase Shift: } PS = \frac{\phi}{W}$$

$$0 = \frac{\phi}{1}$$

$$0 = \phi$$

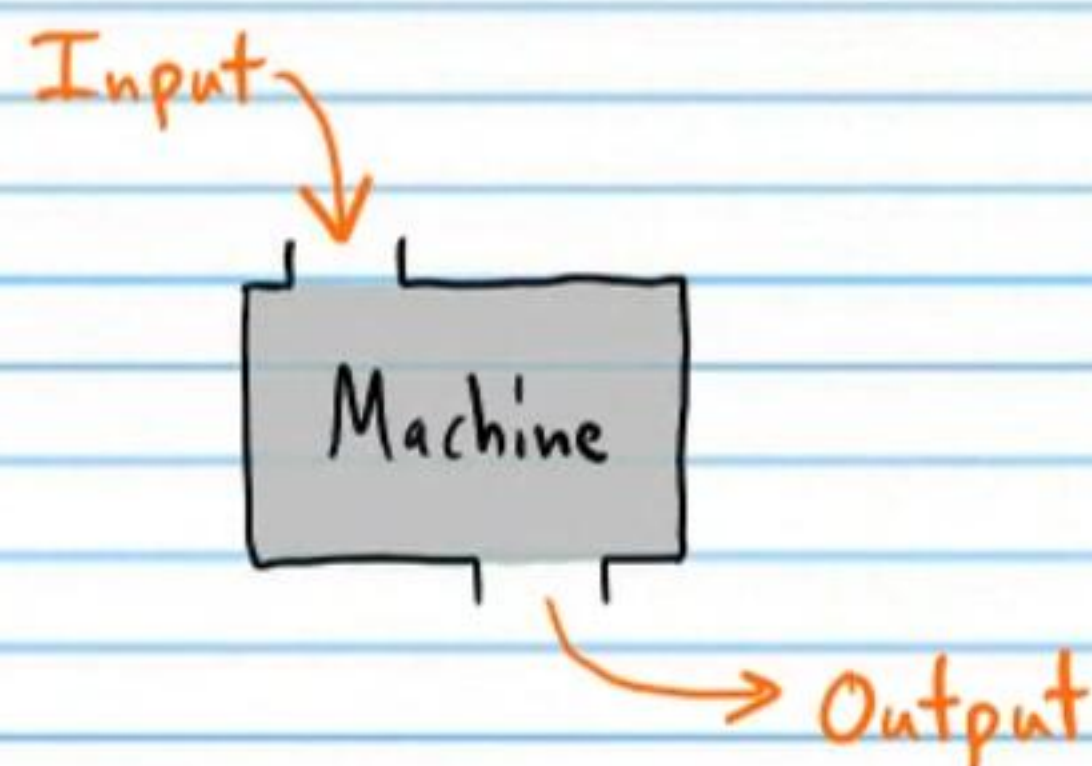


$$y = (1) \tan(1 \cdot x - 0) + (-2)$$

$$\boxed{y = \tan x - 2}$$

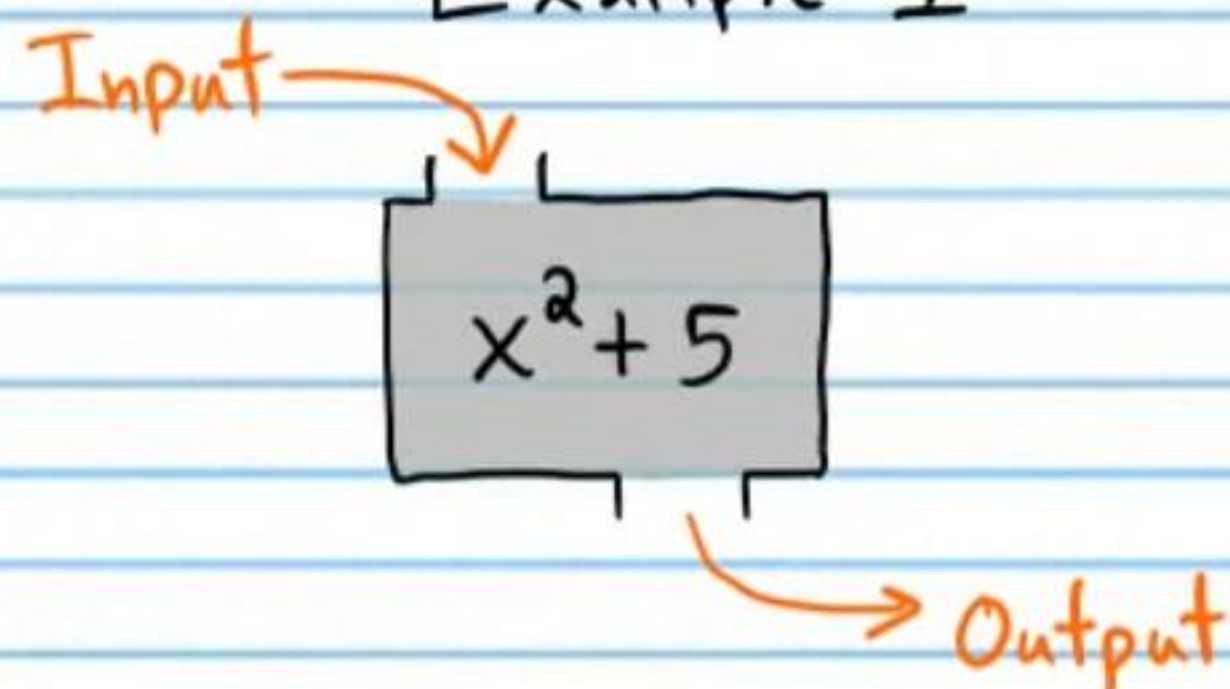
The General Concept of a Function

A function can be thought of as a machine.

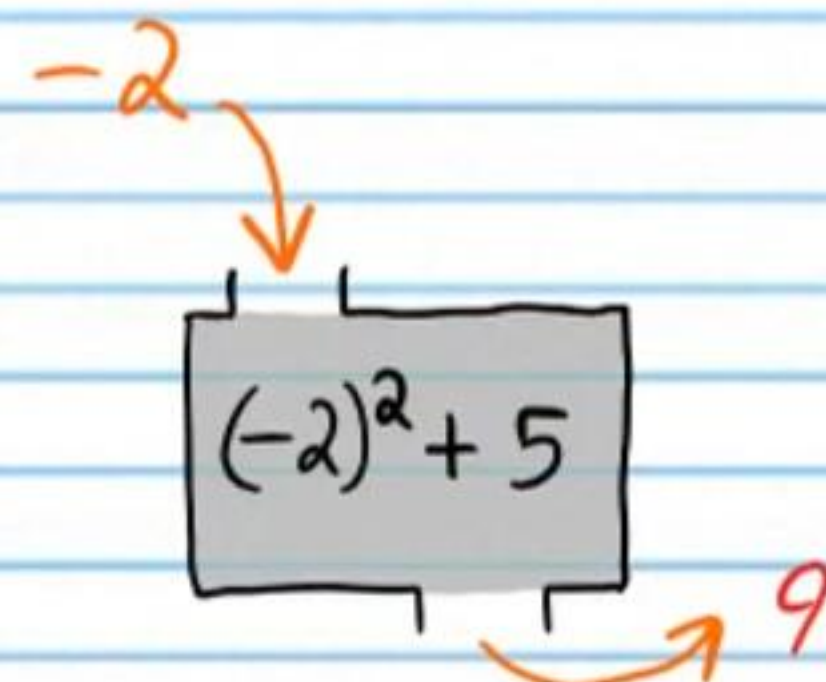
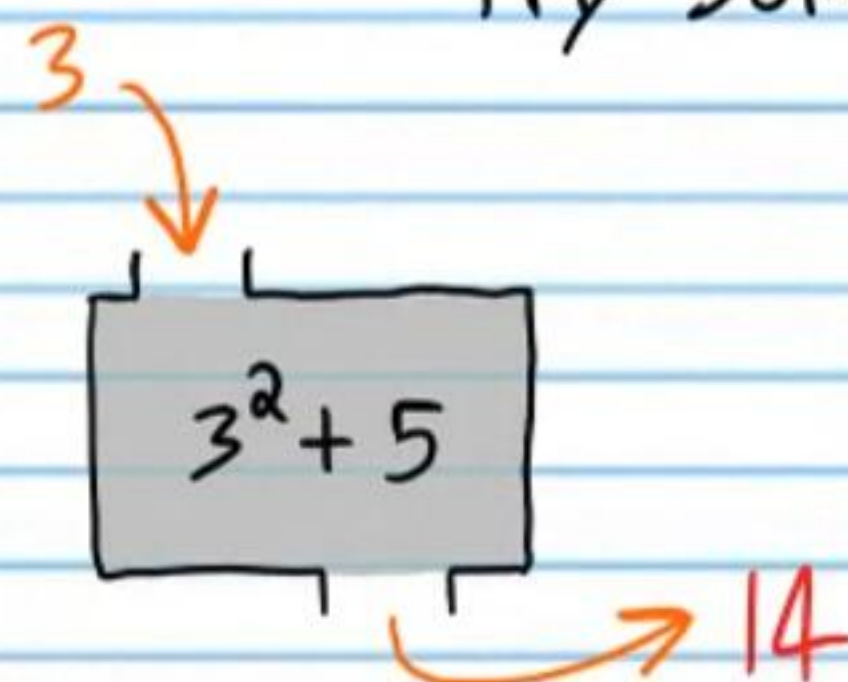


In the subject of mathematics, a function is a set of operations that are performed on a number.

Example I



Try Some Inputs



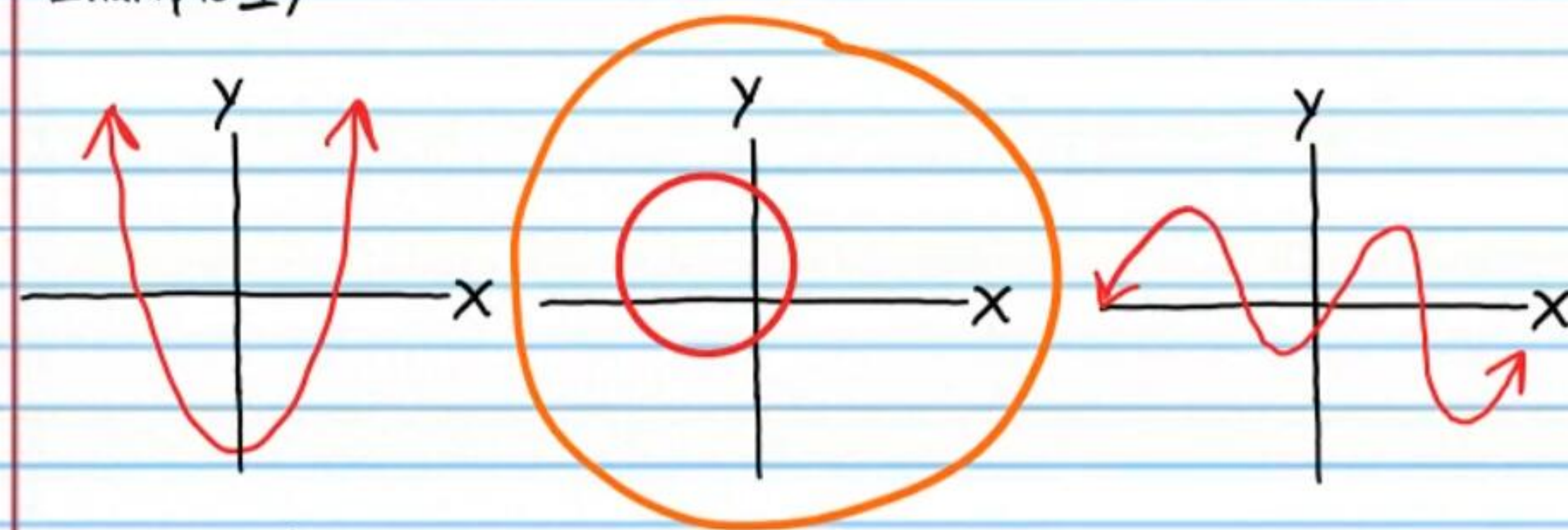
A Necessary Characteristic of Functions

In order for an expression to qualify as a function, every input must produce **ONLY ONE** output.

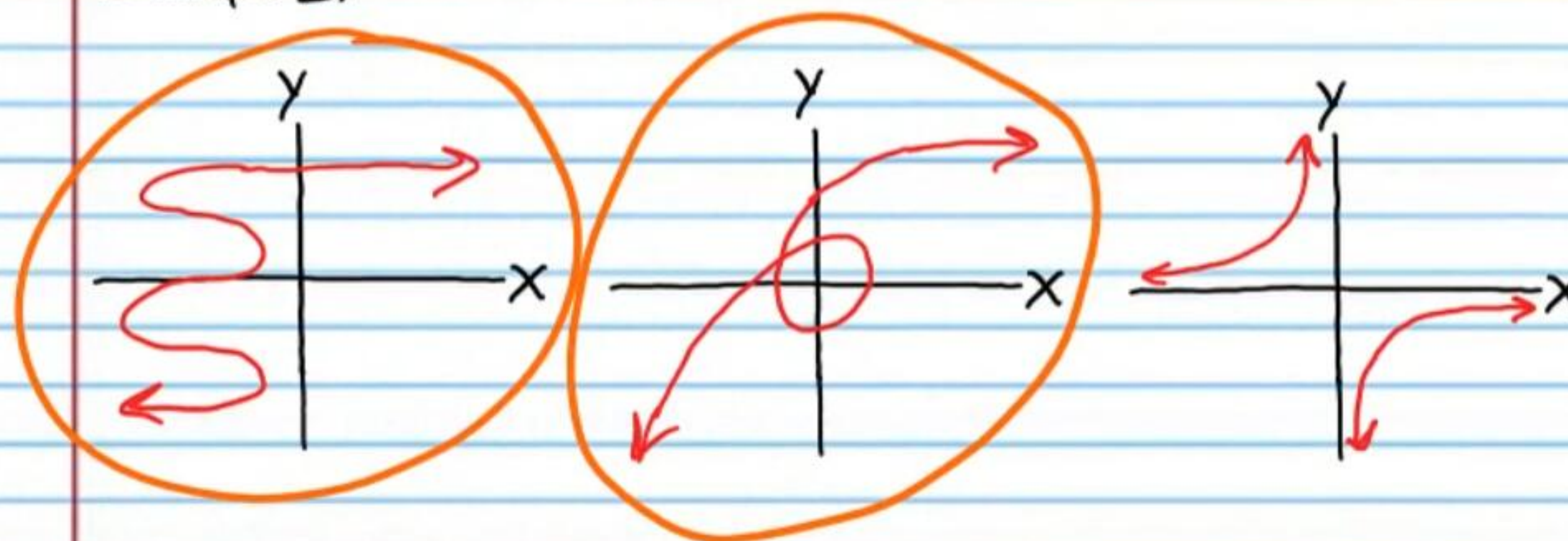
The Vertical Line Test: If any vertical line touches a graph more than once, then the graph cannot be a function.

Circle the relations below that cannot be functions.

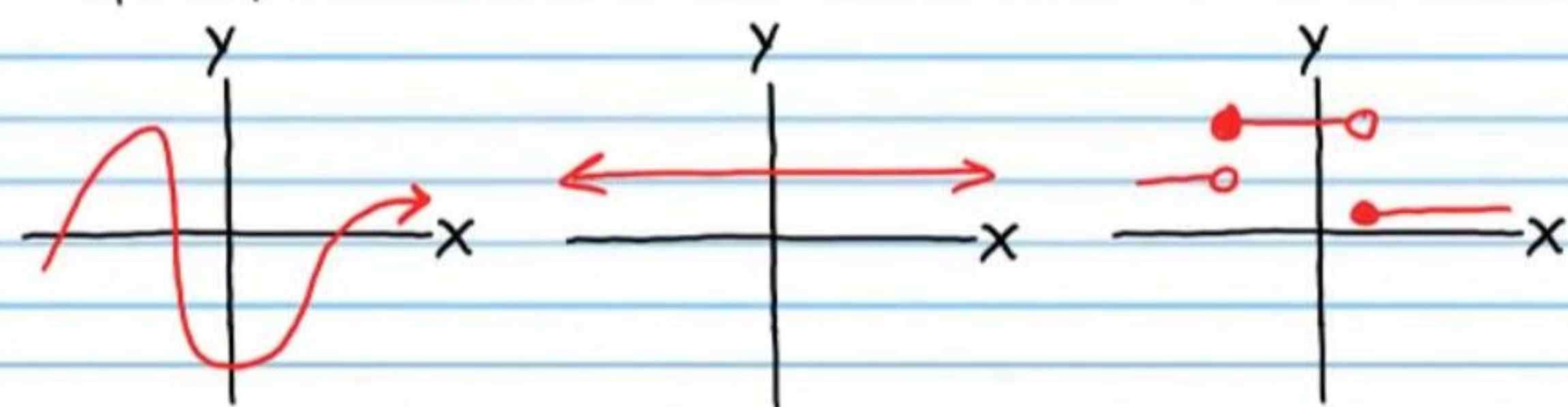
• Example I)



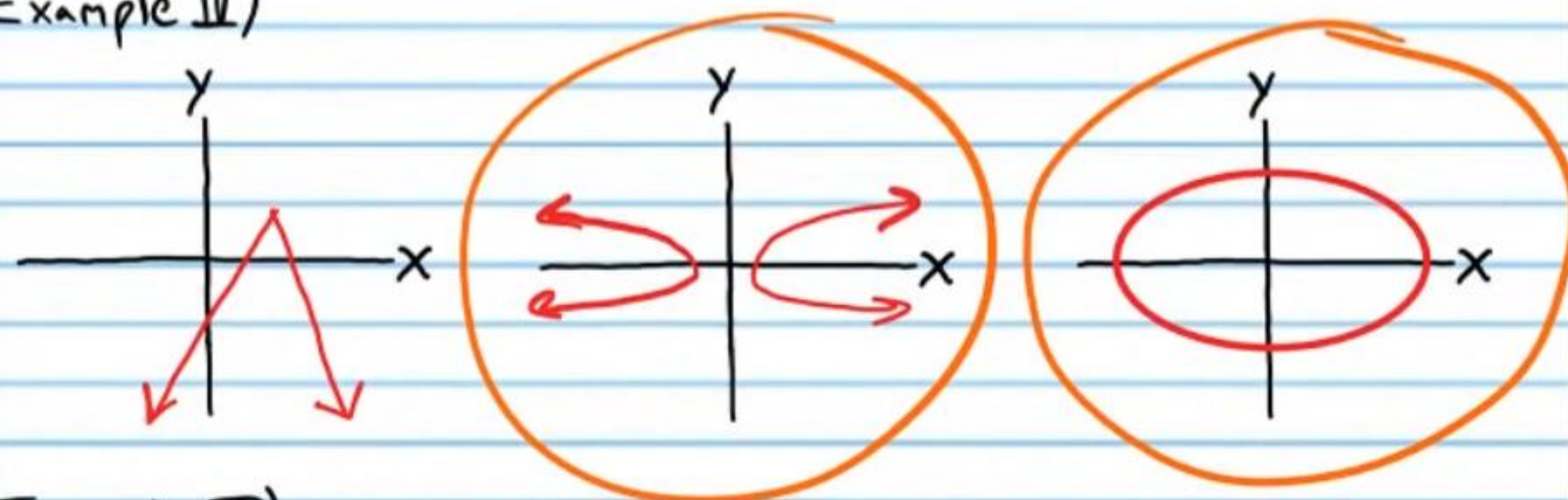
Example II)



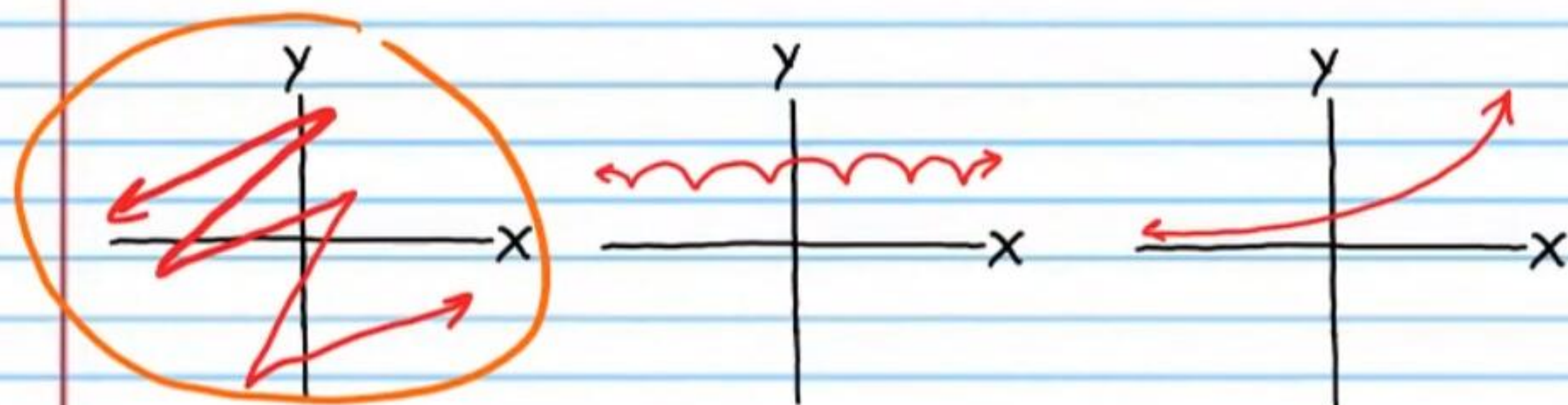
Example III)



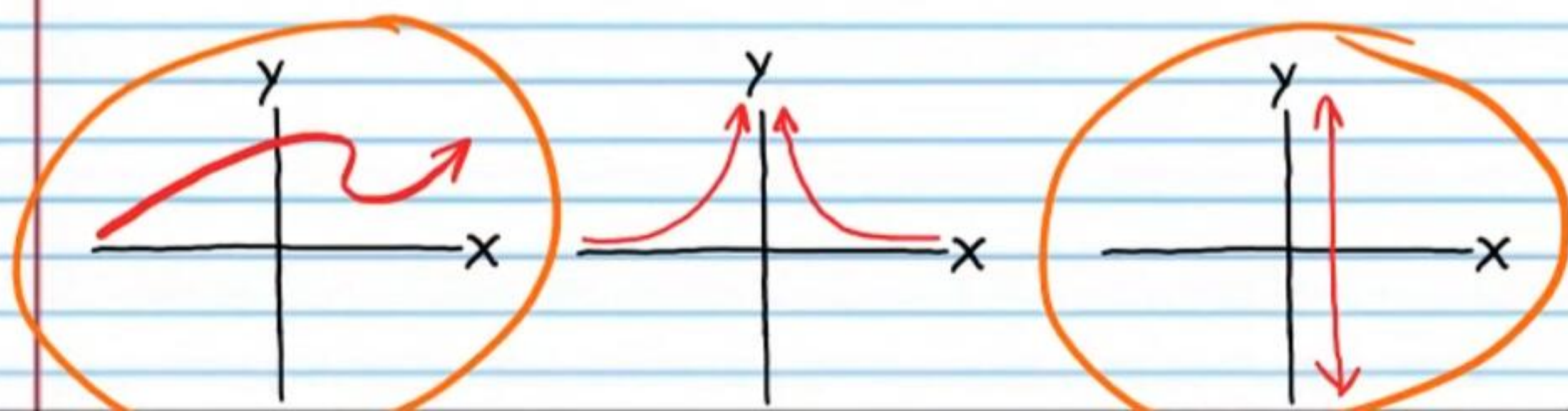
• Example IV)



Example V)



Example VI)



The Inverse of a Function

The Inverse of a Function: A function that performs operations that are the reverse of the original function.

* Warning: The inverse of a function $f(x)$ is notated $f^{-1}(x)$, but **NEGATIVE ONE IN THIS NOTATION IS NOT AN EXPONENT!**

Example: Find the inverse of the function $f(x) = 2x + 3$.

Forward (i.e. the original function)

i) Start x ii) Multiply by 2 $2x$ iii) Add 3 $2x + 3$

Reverse (i.e. the inverse function)

iii) Divide by 2 $\frac{x-3}{2}$ ii) Subtract 3 $x-3$ i) Start x

Inverse Function: $f^{-1}(x) = \frac{x-3}{2}$

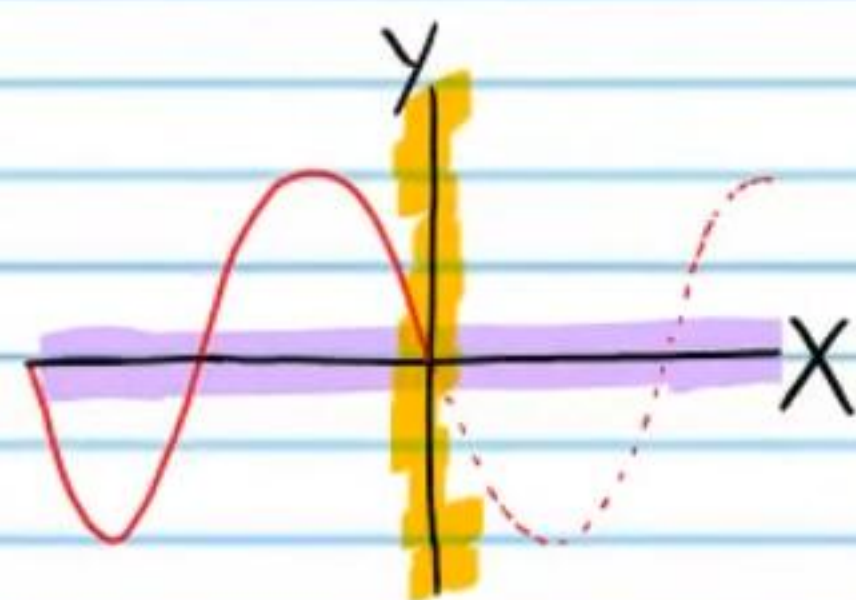
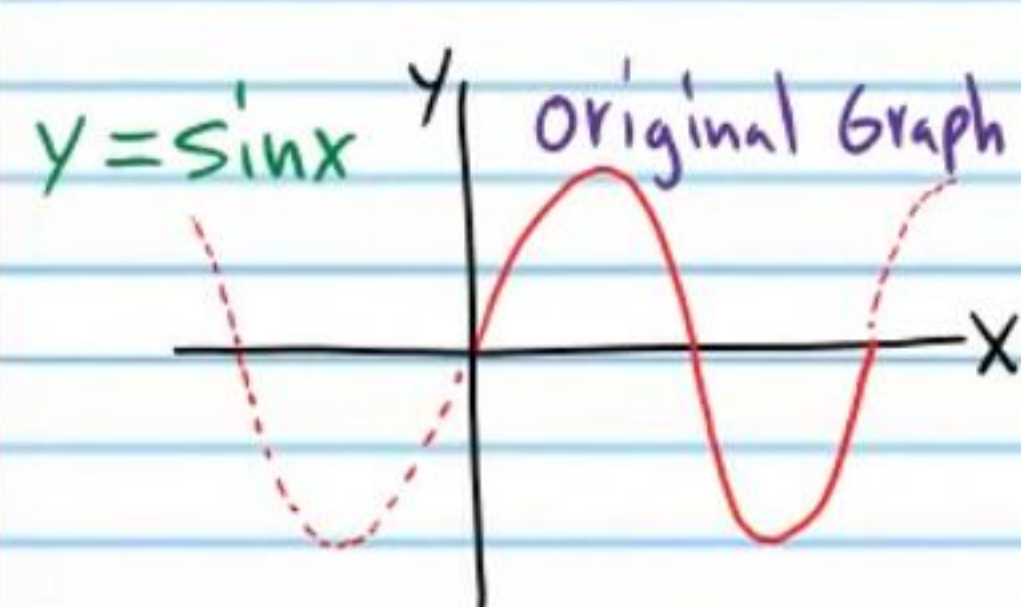
Drawing the Graph of An Inverse Function

To find the graph of an inverse function,

- I) Reflect the original graph across the y -axis.
- II) Then rotate 90° clockwise.
- III) Restrict the range of the relation so that it can qualify as a function.

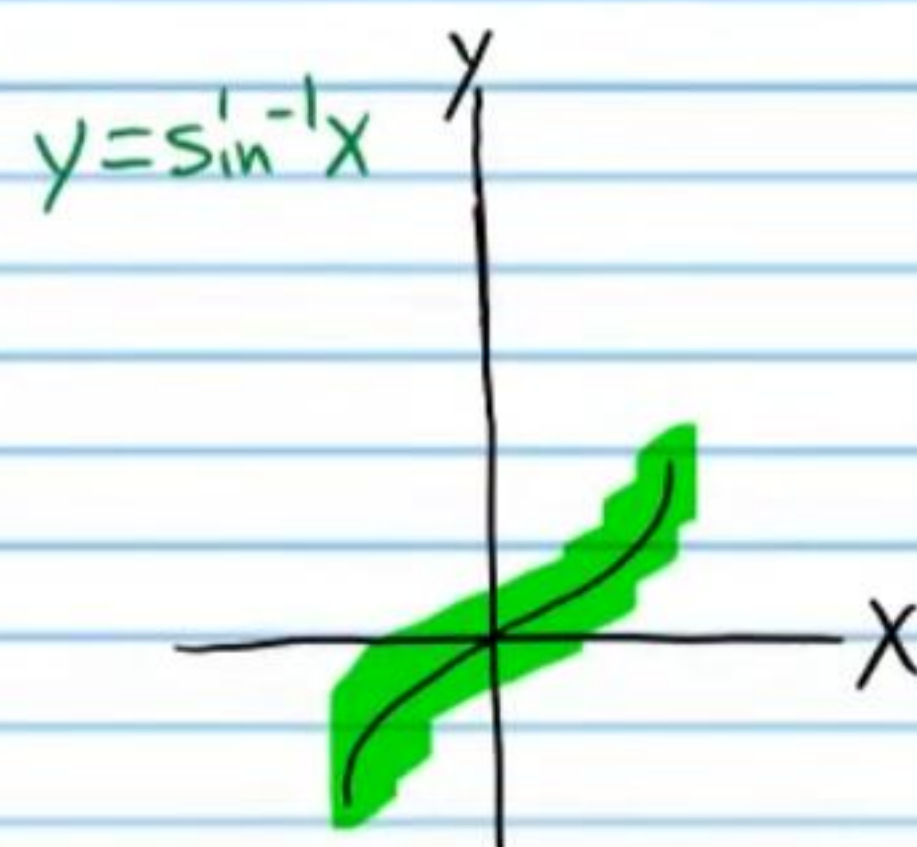
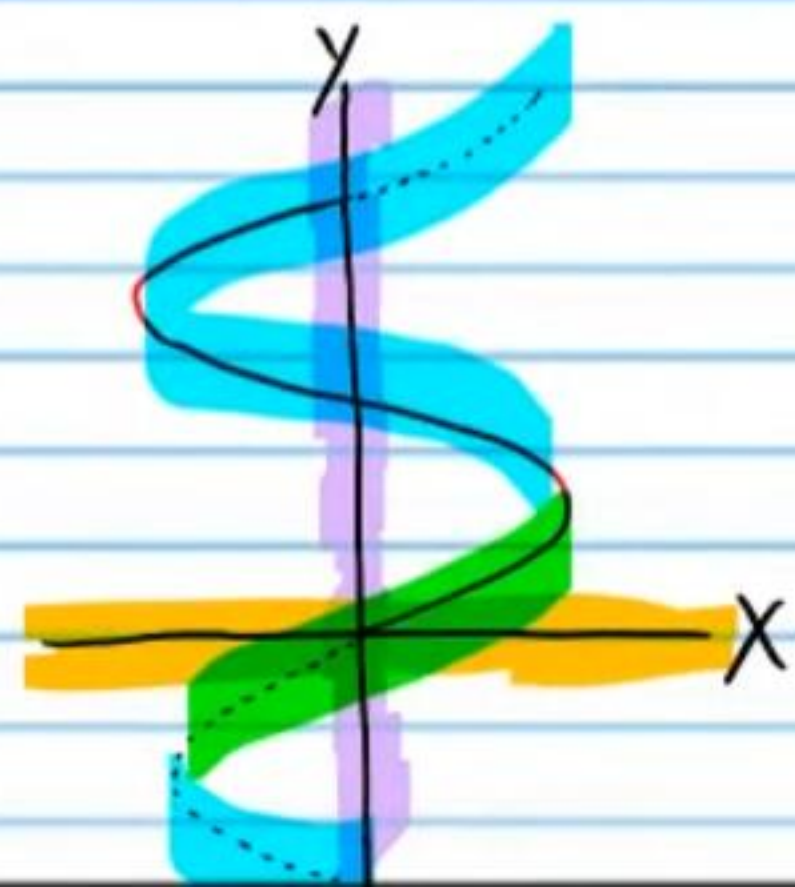
Example:

I) Reflect across the y -axis.



II) Rotate 90° clockwise.

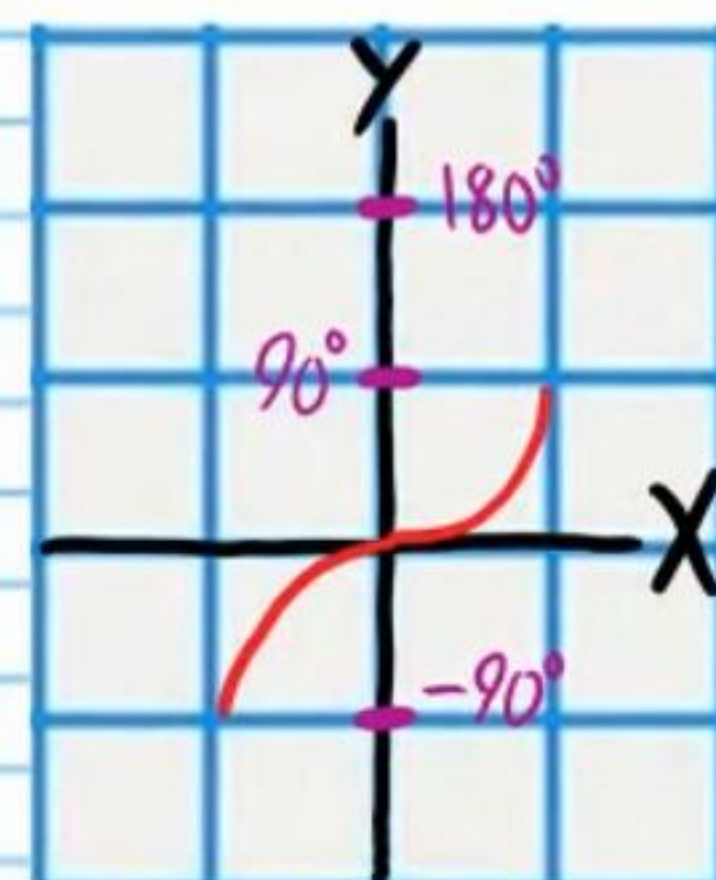
III) Restrict the range so that the corresponding equation can qualify as a function.



Inverse Trigonometric Functions and Their Graphs

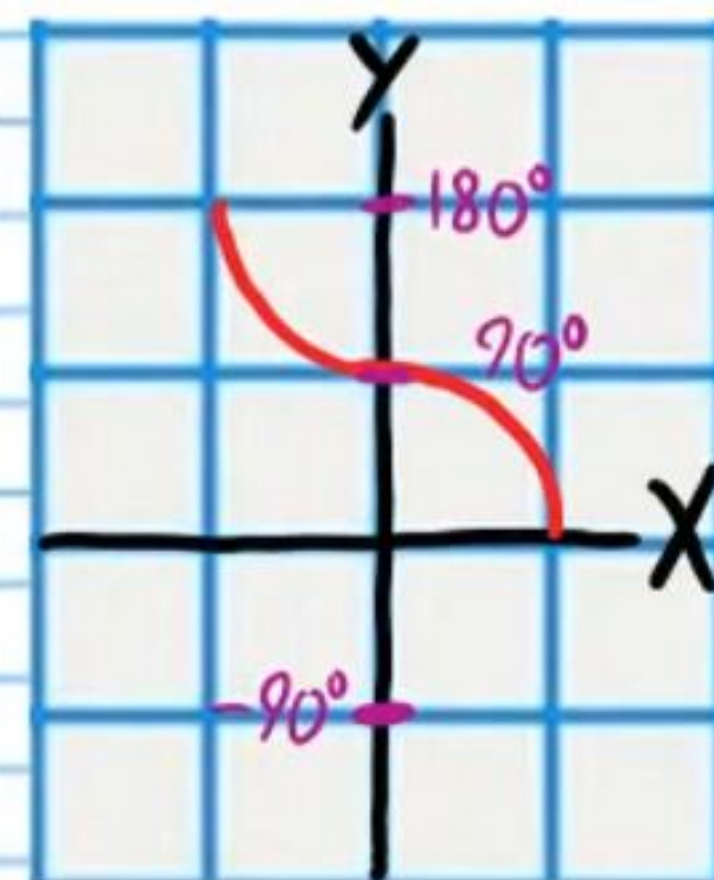
$$y = \sin^{-1} x$$

$$\text{Range: } -90^\circ \leq y \leq 90^\circ$$



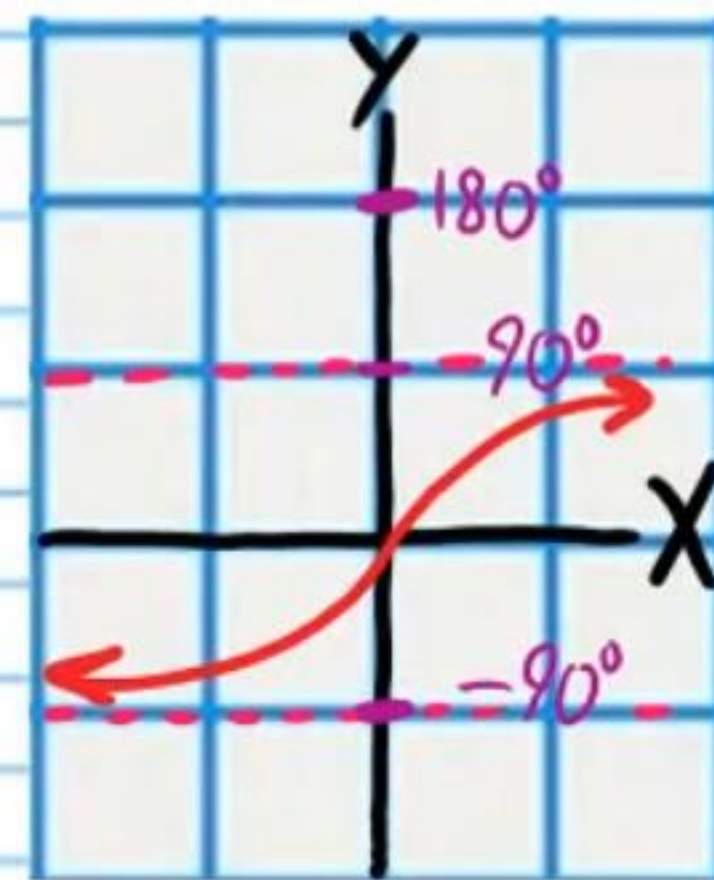
$$y = \cos^{-1} x$$

$$0^\circ \leq y \leq 180^\circ$$



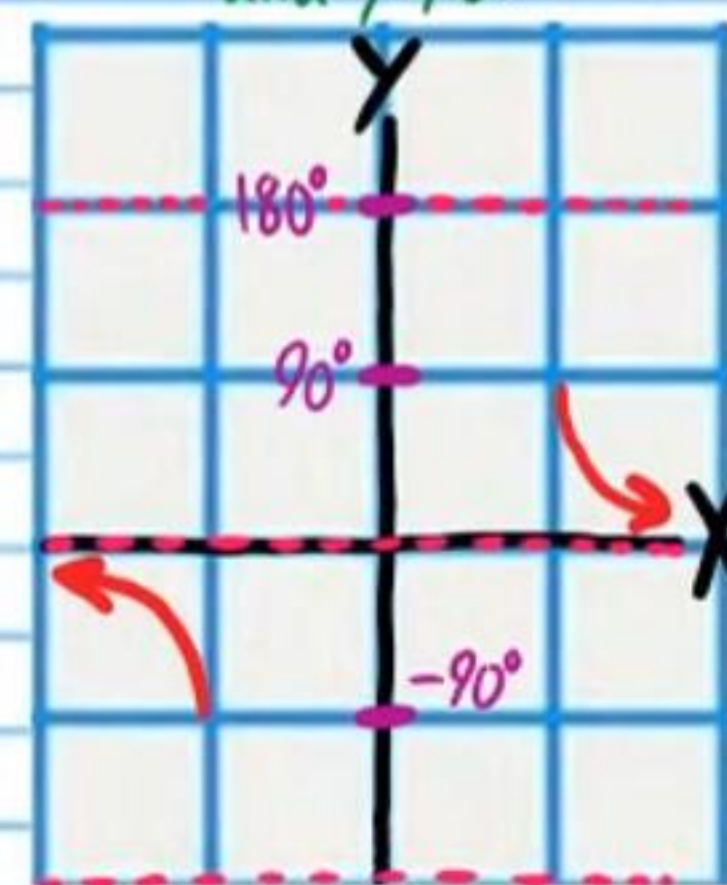
$$y = \tan^{-1} x$$

$$-90^\circ < y < 90^\circ$$



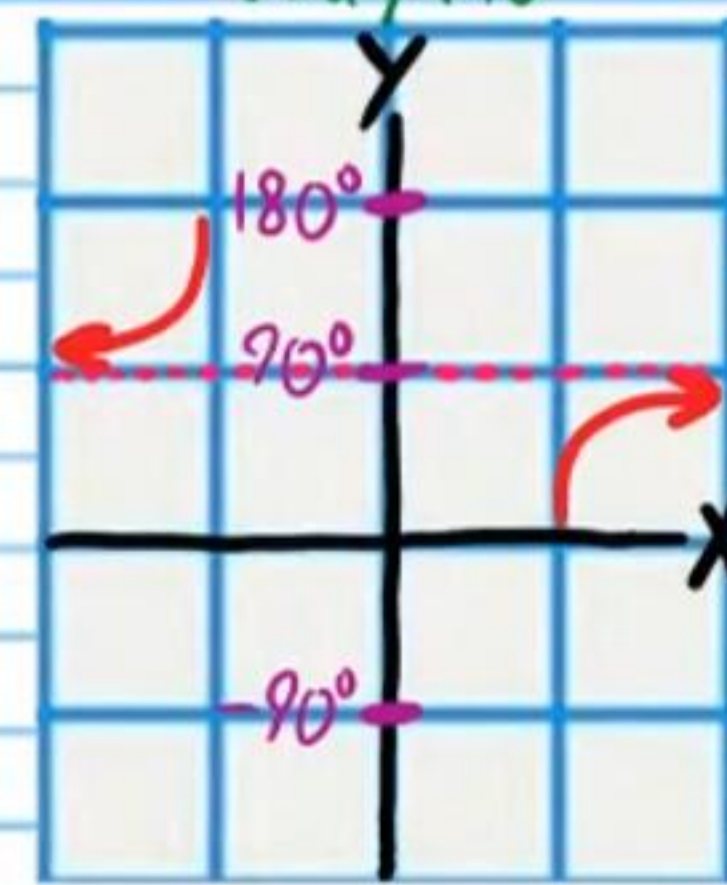
$$y = \csc^{-1} x$$

$$\text{Range: } -90^\circ \leq y \leq 90^\circ \text{ and } y \neq 0$$



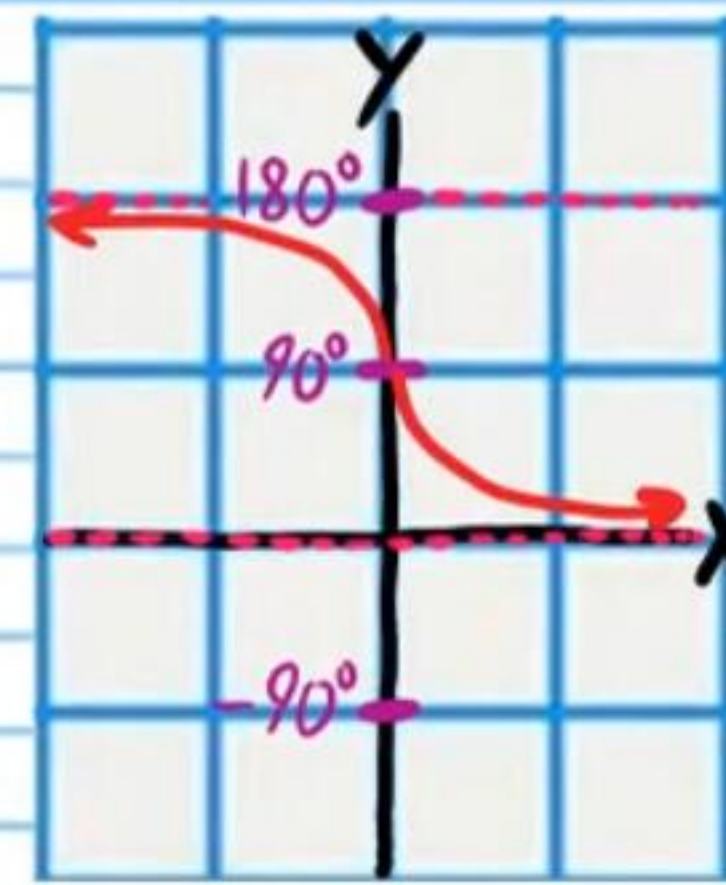
$$y = \sec^{-1} x$$

$$0^\circ \leq y \leq 180^\circ \text{ and } y \neq 90^\circ$$



$$y = \cot^{-1} x$$

$$0^\circ < y < 180^\circ$$



Finding the Values of Inverse Trigonometric Expressions

Find the values of the following expressions. *You will need a calculator.*

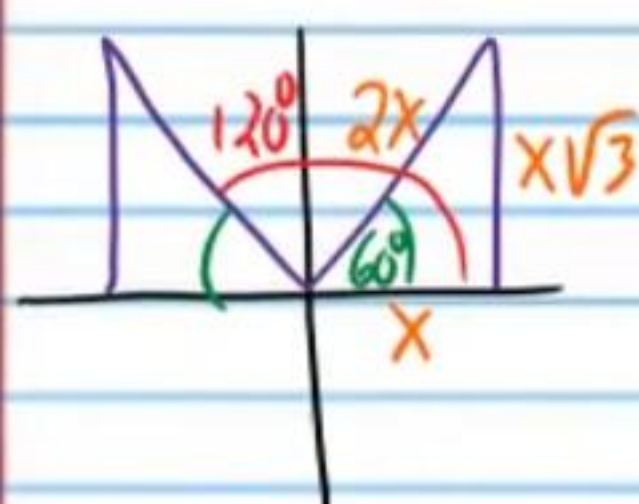
- ⑦ $\cos^{-1}(-0.1) \approx \boxed{95.74^\circ}$
- ⑧ $\tan^{-1}(2.43) \approx \boxed{67.63^\circ}$
- ⑨ $\sin^{-1}(0.4) \approx \boxed{23.58^\circ}$
- ⑩ $\cos^{-1}(0.733) \approx \boxed{42.86^\circ}$

Find the values of the following expressions *without a calculator.*

⑪ $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\sin \theta = \frac{\sqrt{3}}{2}$



$\theta = 60^\circ, 120^\circ$

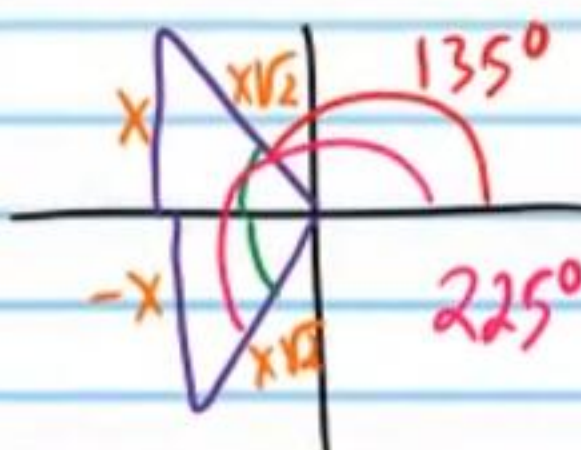
$\theta = 60^\circ$

$\theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \boxed{60^\circ}$

⑫ $\sec^{-1}(-\sqrt{2})$

$\theta = \sec^{-1}(-\sqrt{2})$

$\sec \theta = -\sqrt{2}$



$\theta = 135^\circ, 225^\circ$

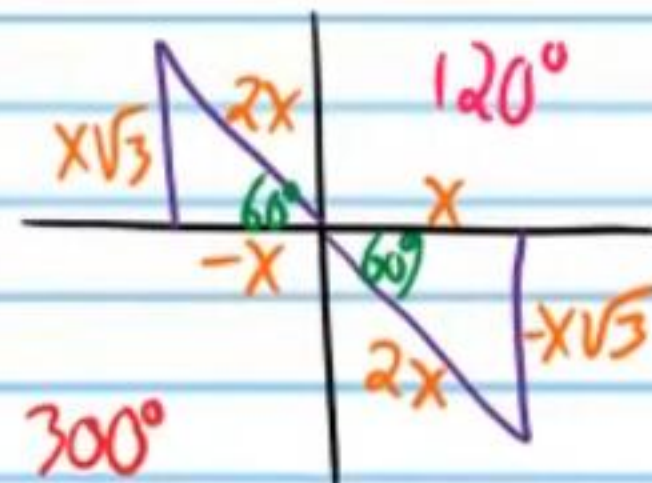
$\theta = 135^\circ$

$\theta = \sec^{-1}(-\sqrt{2}) = \boxed{135^\circ}$

⑬ $\tan^{-1}(-\sqrt{3})$

$\theta = \tan^{-1}(-\sqrt{3})$

$\tan \theta = -\sqrt{3}$



$\theta = 120^\circ, 300^\circ$

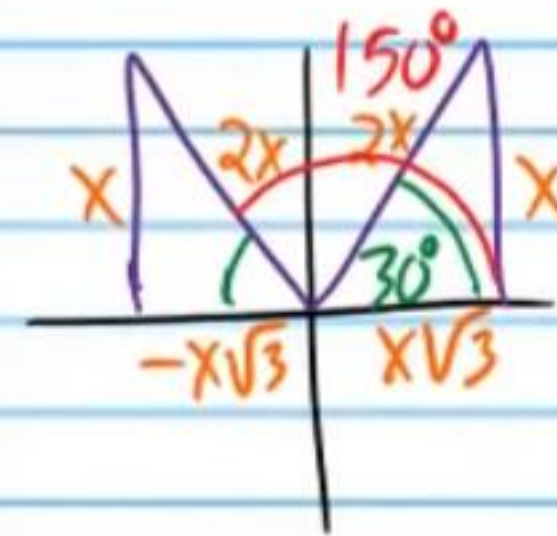
$\theta = -60^\circ$

$\theta = \tan^{-1}(-\sqrt{3}) = \boxed{-60^\circ}$

⑭ $\csc^{-1}(2)$

$\theta = \csc^{-1}(2)$

$\csc \theta = 2$



$\theta = 30^\circ, 150^\circ$

$\theta = \csc^{-1}(2) = \boxed{30^\circ}$

⑮ $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\theta = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\cos \theta = \frac{\sqrt{3}}{2}$

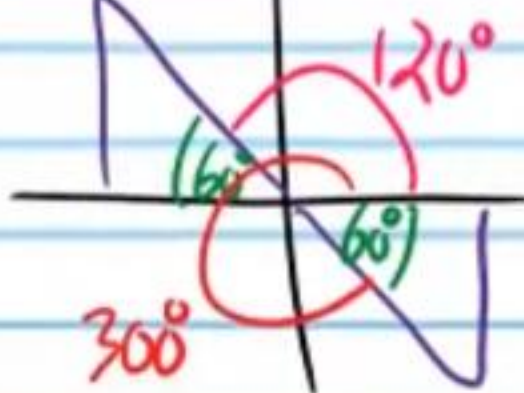


$\theta = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \boxed{30^\circ}$

⑯ $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

$\theta = \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

$\cot \theta = -\frac{1}{\sqrt{3}}$

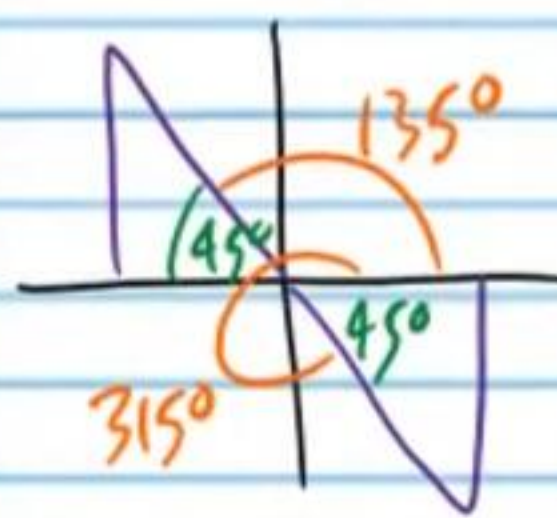


$\theta = \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \boxed{120^\circ}$

⑰ $\tan^{-1}(-1)$

$\theta = \tan^{-1}(-1)$

$\tan \theta = -1$

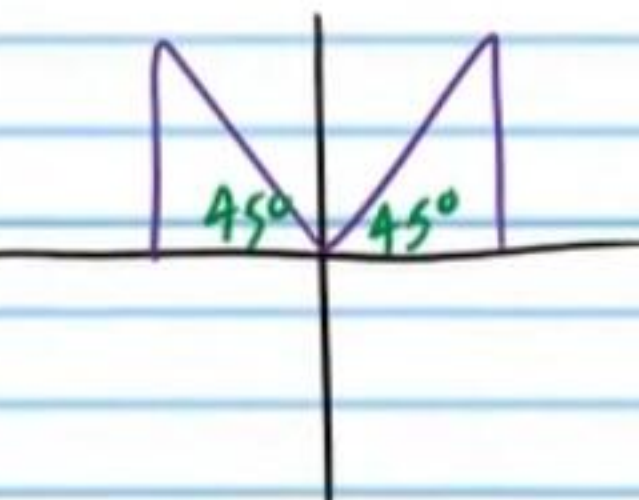


$\theta = \tan^{-1}(-1) = \boxed{-45^\circ}$

⑱ $\csc^{-1}(\sqrt{2})$

$\theta = \csc^{-1}(\sqrt{2})$

$\csc \theta = \sqrt{2}$



$\theta = \csc^{-1}(\sqrt{2}) = \boxed{45^\circ}$

⑲ $\cos^{-1}(1)$

$\theta = \cos^{-1}(1)$

$\cos \theta = 1$

$\theta = 0^\circ$

$\theta = \cos^{-1}(1) = \boxed{0^\circ}$

②① $\cot^{-1}(0)$

$\theta = \cot^{-1}(0)$

$\cot \theta = 0$

$\theta = 90^\circ$

$\theta = \cot^{-1}(0) = \boxed{90^\circ}$

★ ②① $\csc^{-1}(0)$

$\theta = \csc^{-1}(0)$

$\csc \theta \neq 0$

Undefined

• ②② $\tan^{-1}(0)$

$\theta = \tan^{-1}(0)$

$\tan \theta = 0$

$\theta = 0^\circ$

$\theta = \tan^{-1}(0) = \boxed{0^\circ}$

★ ②③ $\sec^{-1}(0)$

$\theta = \sec^{-1}(0)$

$\sec \theta \neq 0$

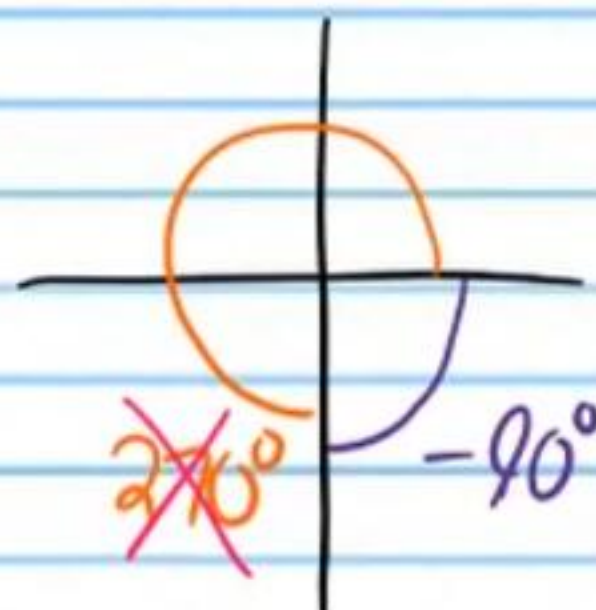
Undefined

②④ $\sin^{-1}(-1)$

$\theta = \sin^{-1}(-1)$

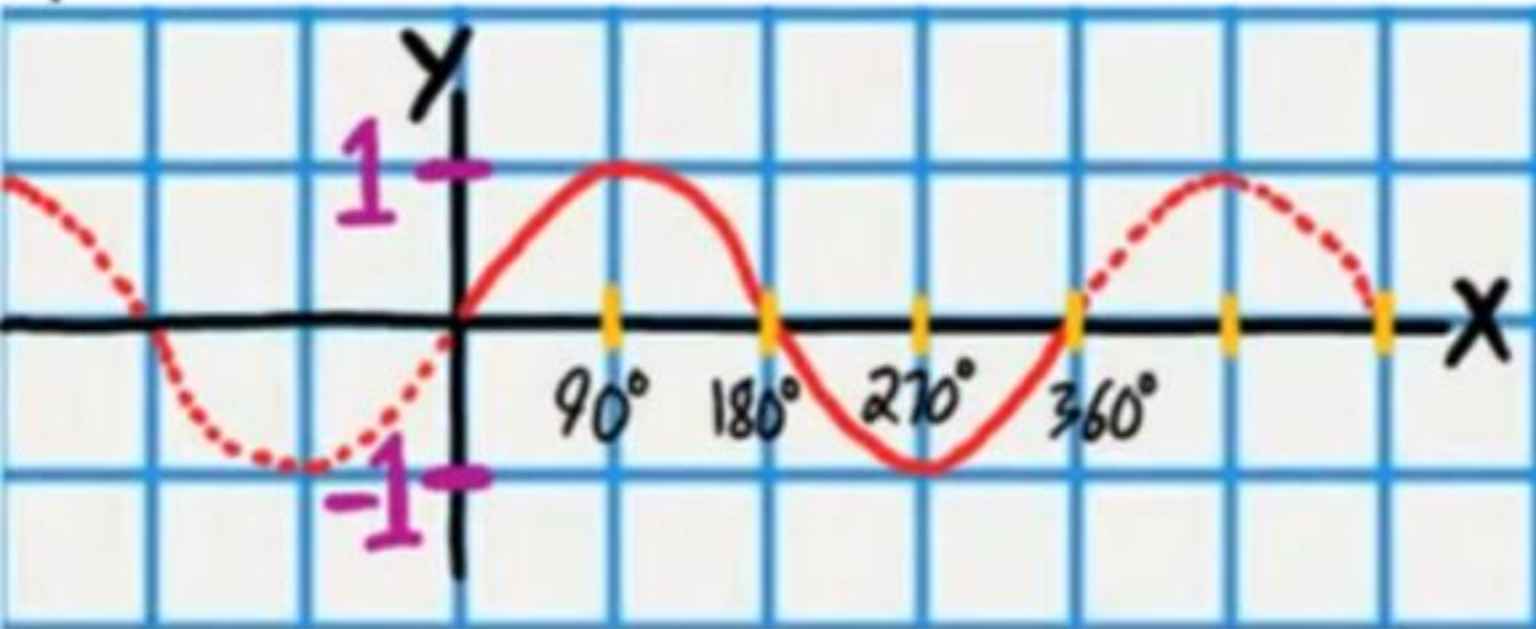
$\sin \theta = -1$

$\theta = 270^\circ$

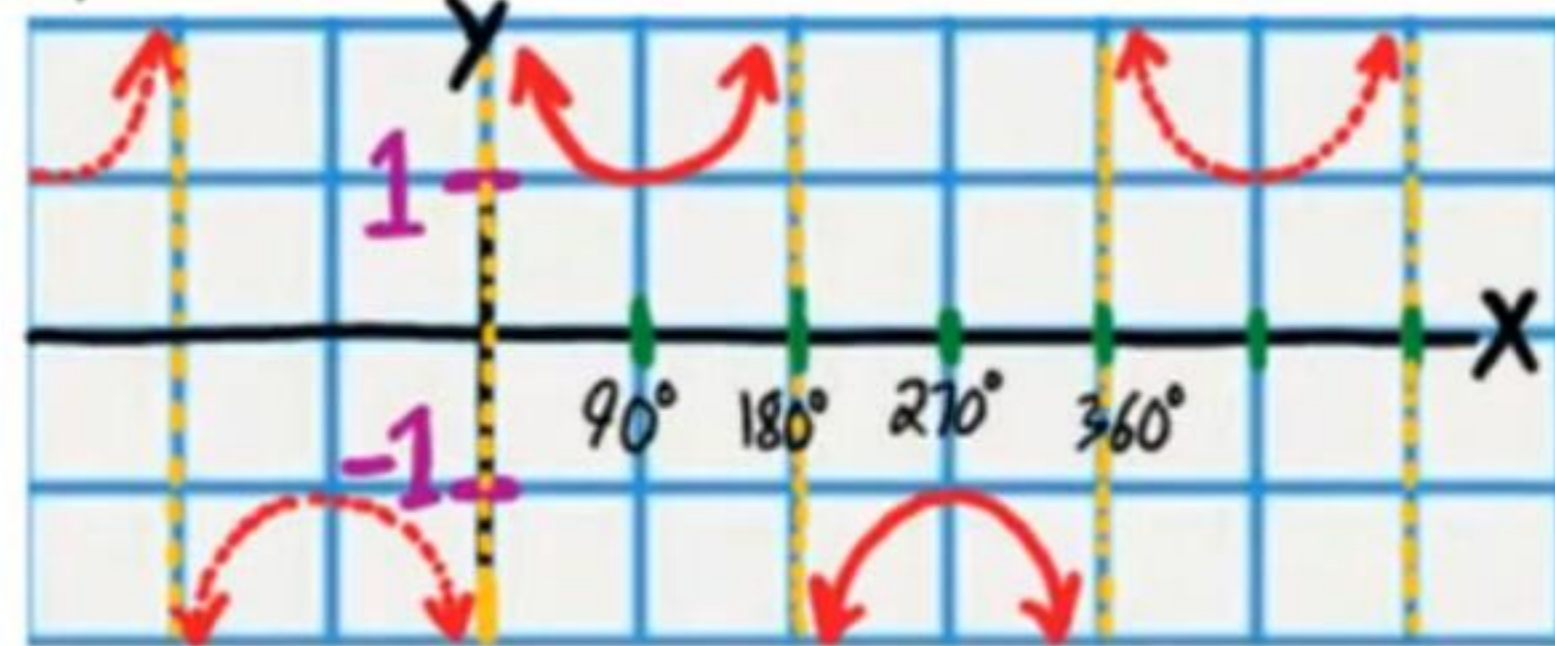


$\theta = \sin^{-1}(-1) = \boxed{-90^\circ}$

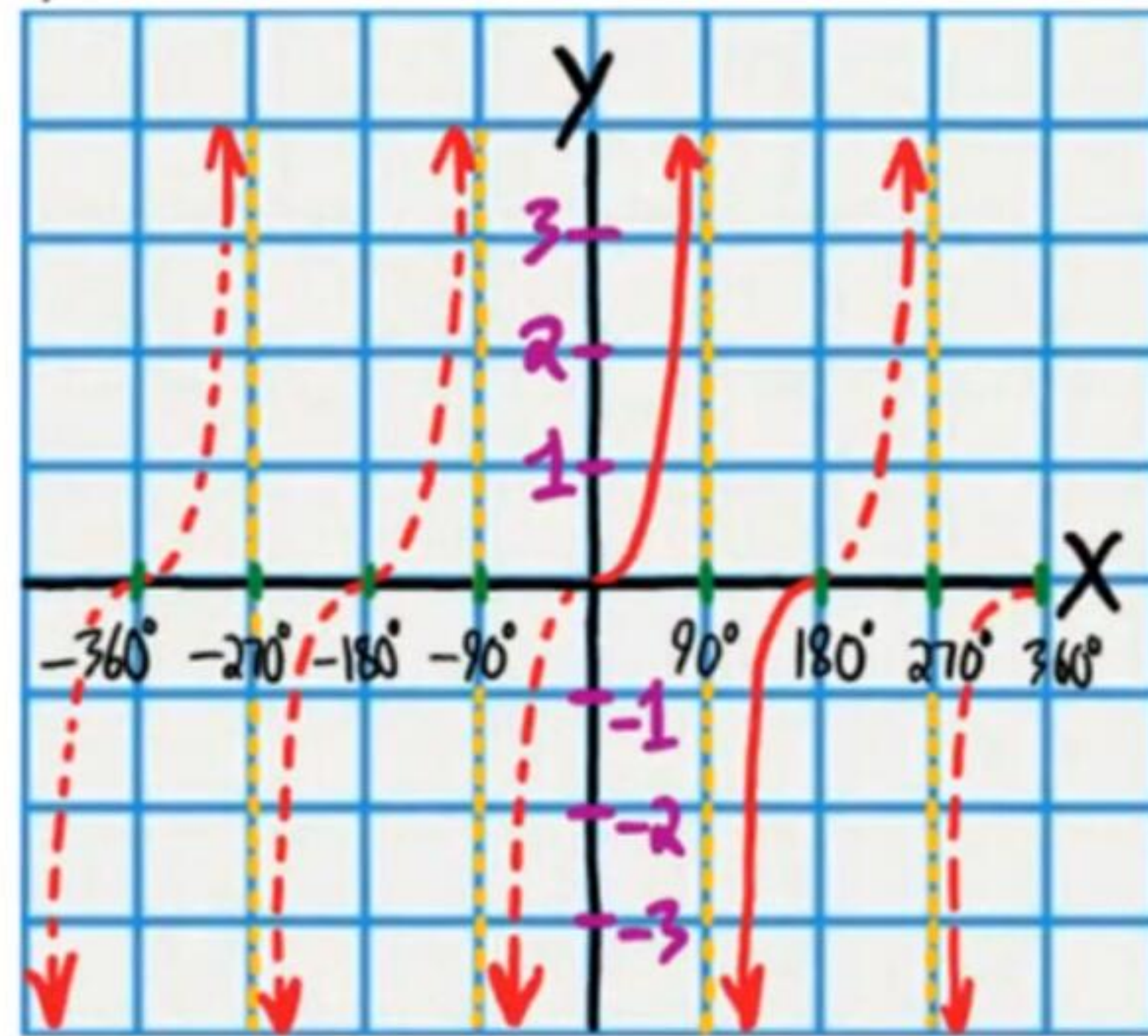
$$y = \sin x$$



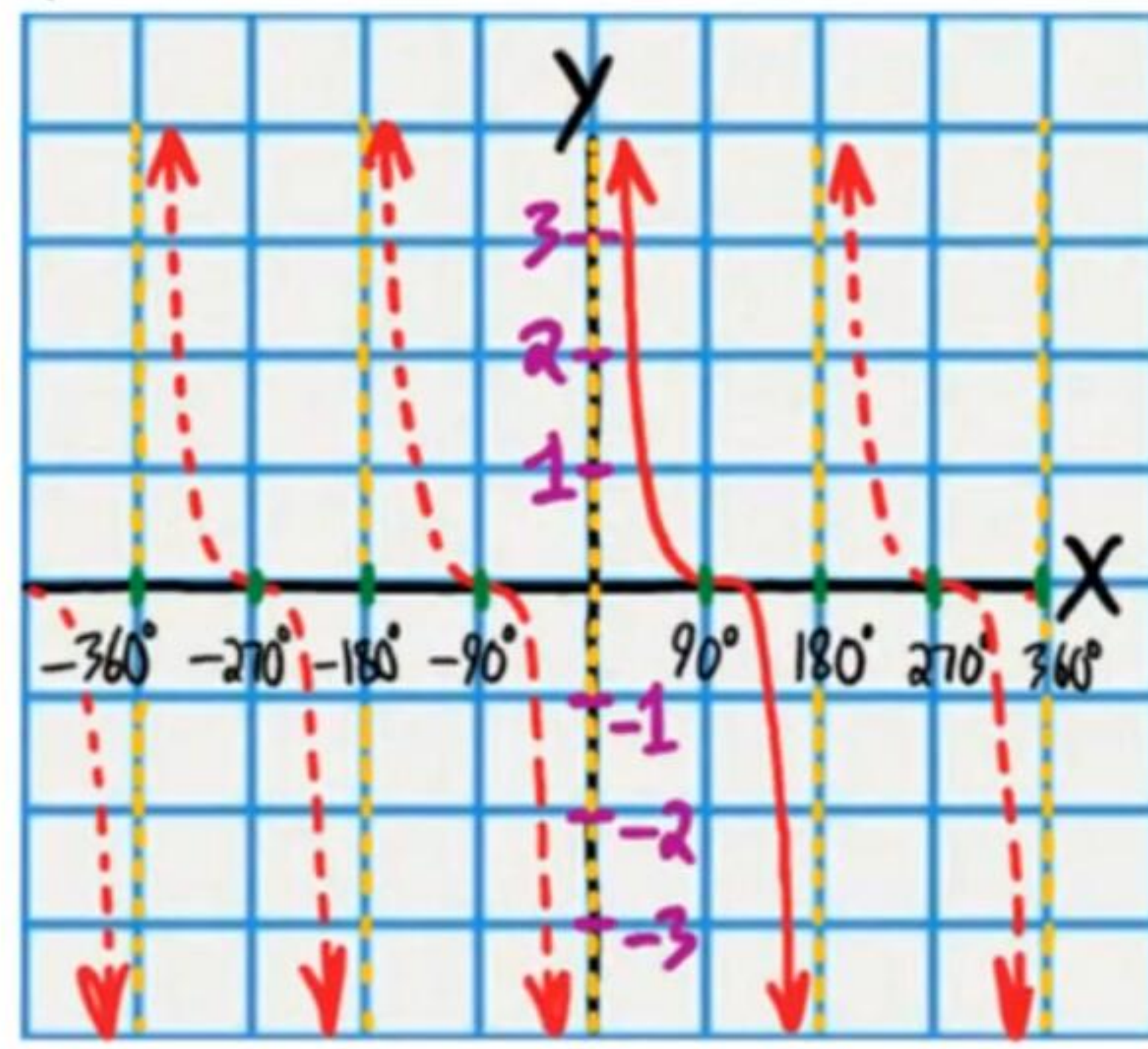
$$y = \csc x$$



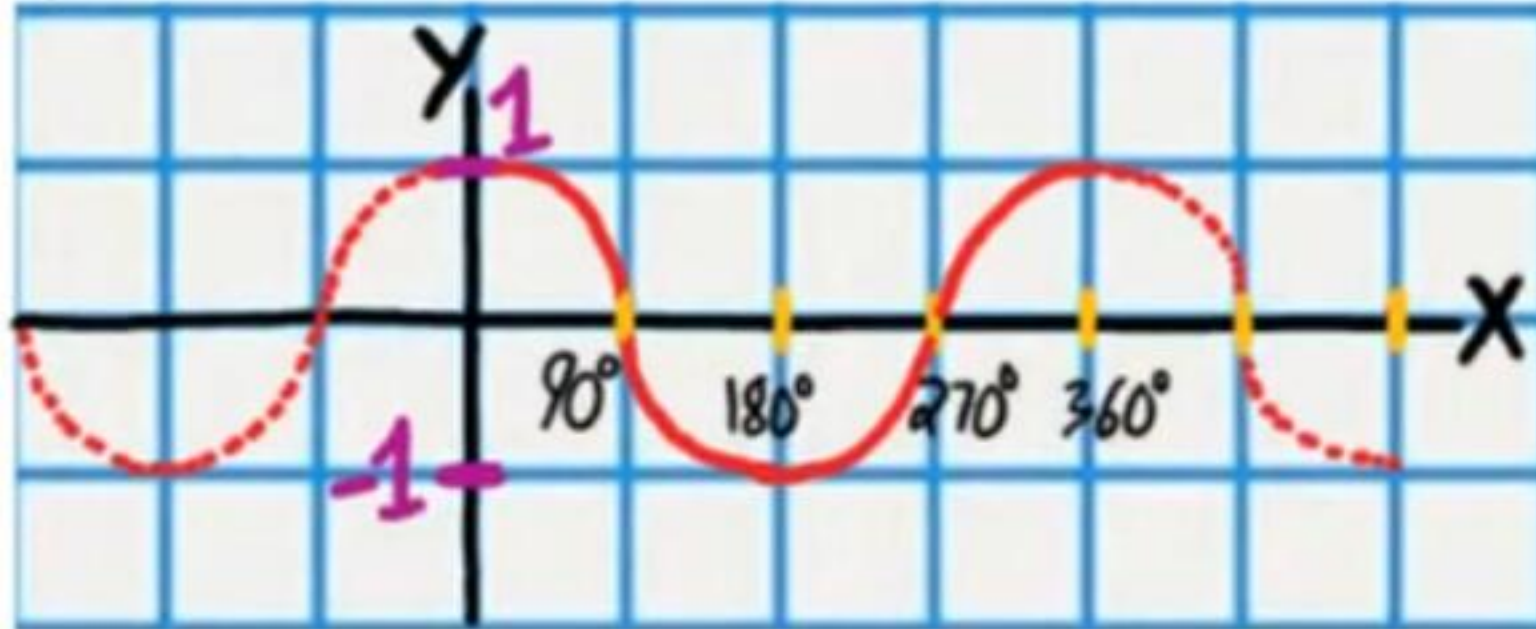
$$y = \tan x$$



$$y = \cot x$$



$$y = \cos x$$



$$y = \sec x$$

